

Morning Paper Report - Tuesday, June 23, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Working Memory

Bradley R. Postle, Klaus Oberauer

The Oxford Handbook of Human Memory, Two Volume Pack, 2024

Source: Crossref; matched terms: working memory, computational model, computational models, human memory

Link: <https://doi.org/10.1093/oxfordhb/9780190917982.013.14>

Goal: Abstract Working memory supports the guidance of current thoughts and actions with temporarily selected representations deriving from either long-term memory or information recently perceived in the environment

Method: From this perspective, the chapter considers the current state of theoretical and computational models, including the distinction between memory systems and state-dependent models, differing accounts of capacity limitations, the relation of working memory to long-term memory, and the relation of working memory to attention and cognitive control

Results: The chapter concludes with a consideration of the neural bases of working memory, emphasizing results from lesion and neurophysiological experiments, carried out with humans and nonhuman primates, that are associated with the theoretical and computational considerations discussed in the chapter

Conclusion: Abstract Working memory supports the guidance of current thoughts and actions with temporarily selected representations deriving from either long-term memory or information recently perceived in the environment

2. Effects of Retro-cue Reliability on Visual Working Memory and Attentional Template Efficiency in Visual Search

Authors unavailable

2025

Source: OSF/PsyArXiv; matched terms: visual working memory, visual search, working memory

Link: https://osf.io/preprints/psyarxiv/yzun9_v3/

Goal: In the visual search literature, an attentional template refers to a mental representation of target features stored in visual working memory

Method: The dual-state model (Olivers et al., 2011) argues that visual working memory can only be focused on a single representation at a time, and therefore, we can only search for one target at a time

Results: The results suggest that participants allocate their resources based on the reliability of the retro-cue and therefore favor the resource hypothesis

Conclusion: The dual-state model (Olivers et al., 2011) argues that visual working memory can only be focused on a single representation at a time, and therefore, we can only search for one target at a time

3. Upgrading the JANET neural network by introducing a new storage buffer of working memory

Antonio Tolic, Biljana Mileva Boshkoska, Sandro Skansi

Neural Network World, 2023

Source: Crossref; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://doi.org/10.14311/nnw.2023.33.024>

Goal: Recurrent neural networks (RNNs), along with long short-term memory networks (LSTMs), have been successfully used on a wide range of sequential data problems and have been entitled as extraordinarily powerful tools for learning and processing such data

Method: However, the search for a new or derived architecture that would model very long-term dependencies is still an active area of research

Results: The proposed architecture has achieved world-class results and it outperforms JANET on multiple benchmark datasets

Conclusion: The proposed architecture has achieved world-class results and it outperforms JANET on multiple benchmark datasets Moreover, the new architecture is applicable to a wider class of problems and showed superior resilience when processing longer sequences, as opposed to JANET which experienced catastrophic failures on certain tasks.

4. Trained recurrent neural networks develop phase-locked limit cycles in a working memory task.

Pals M, Macke JH, Barak O
PLoS computational biology, 2024

Source: PubMed; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://pubmed.ncbi.nlm.nih.gov/38315736/>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

5. Pruning recurrent neural networks replicates adolescent changes in working memory and reinforcement learning.

Averbeck BB
Proceedings of the National Academy of Sciences of the United States of America, 2022

Source: PubMed; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://pubmed.ncbi.nlm.nih.gov/35622896/>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.